Airbnb Data Analysis Report – Chicago & New Orleans

1. Project Title

Transforming EDA Projects into Power BI Dashboard for Airbnb Listings: A Comparative Study of Chicago and New Orleans

2. Problem Statement

Airbnb, a global platform for short-term property rentals, enables users to book accommodations in diverse urban areas. This study aims to explore Airbnb operations in **Chicago** and **New Orleans** to identify trends, differences, and patterns in listings, pricing, property types, and host behavior.

The goal is to use **Power BI** to provide a **comprehensive comparative analysis** that supports actionable insights for Airbnb's management and hosts.

Key Questions:

1. Which neighbourhoods are most popular, and how do their listings and prices differ?
2. What are the dominant room types and property types in each city?
3. How does pricing vary with location, property type, and reviews?
4. How do host types correlate with reviews and pricing?

3. Objective of the Study

- Compare Airbnb listings in **Chicago** and **New Orleans**.
- Identify popular areas, property types, and room types.
- Analyze pricing trends across locations, property types, and reviews.
- Examine host behavior and its influence on pricing and reviews.
- Build a **Power BI dashboard** that visualizes these insights effectively.

4. Dataset Description

Source: Inside Airbnb
Cities: Chicago and New Orleans

Key Attributes:

- id – Unique identifier for each listing
- name – Name of the listing
- host_id – Unique identifier for host
- host_name – Host's name

- neighbourhood_group – Larger area of the city
- neighbourhood – Specific neighbourhood
- latitude, longitude – Coordinates
- room_type – Private room, Entire home/apt, Shared room
- price – Price per night
- minimum_nights – Minimum nights required
- number_of_reviews – Total reviews received
- last_review – Date of last review
- reviews_per_month – Average monthly reviews
- availability_365 – Number of available days per year

Data Cleaning Steps (Performed in Jupyter Notebook):

- Removed duplicates and missing values
- Standardized text (lowercase, trimmed spaces)
- Cleaned price column and converted to numeric
- Converted date fields to datetime format
- Checked for outliers in pricing and availability

5. Methodology

1. **Data Cleaning** – Ensured data consistency, removed duplicates, and standardized columns.
2. **Exploratory Data Analysis (EDA)** – Analyzed neighbourhood popularity, property and room type distribution, pricing trends, and host activity.
3. **Power BI Dashboard** – Built a **4-section dashboard** covering Overview, Property Analysis, Pricing Analysis, and Host Analysis with slicers for city, room type, neighbourhood, and price.
4. **Insights Generation** – Derived actionable insights from patterns observed in the data.

Insights & Findings

6.1 Overview of Airbnb

- **Popular Neighbourhoods:**
    - Chicago: Downtown and Lincoln Park have the most listings
    - New Orleans: French Quarter and Garden District have the most listings
- **Insight:** Chicago has more listings overall, but New Orleans shows slightly higher prices. Central areas in both cities attract more guests.

6.2 Property Analysis

- **Room Types:** Most listings are Entire homes/apartments, followed by Private rooms, with a few Shared rooms.
- **Property Types:** Apartments dominate Chicago, while Houses dominate New Orleans.
- **Insight:** Travelers prefer privacy, and property type distribution varies by city.

6.3 Pricing Analysis

- **Pricing Trends:** Entire homes/apartments are more expensive than Private or Shared rooms. Central neighbourhoods have higher prices than outer areas. Listings with more reviews tend to have slightly higher prices.
- **Insight:** Popular and central locations command higher prices. Highly-rated listings are slightly more expensive than low-reviewed ones.

6.4 Host Analysis

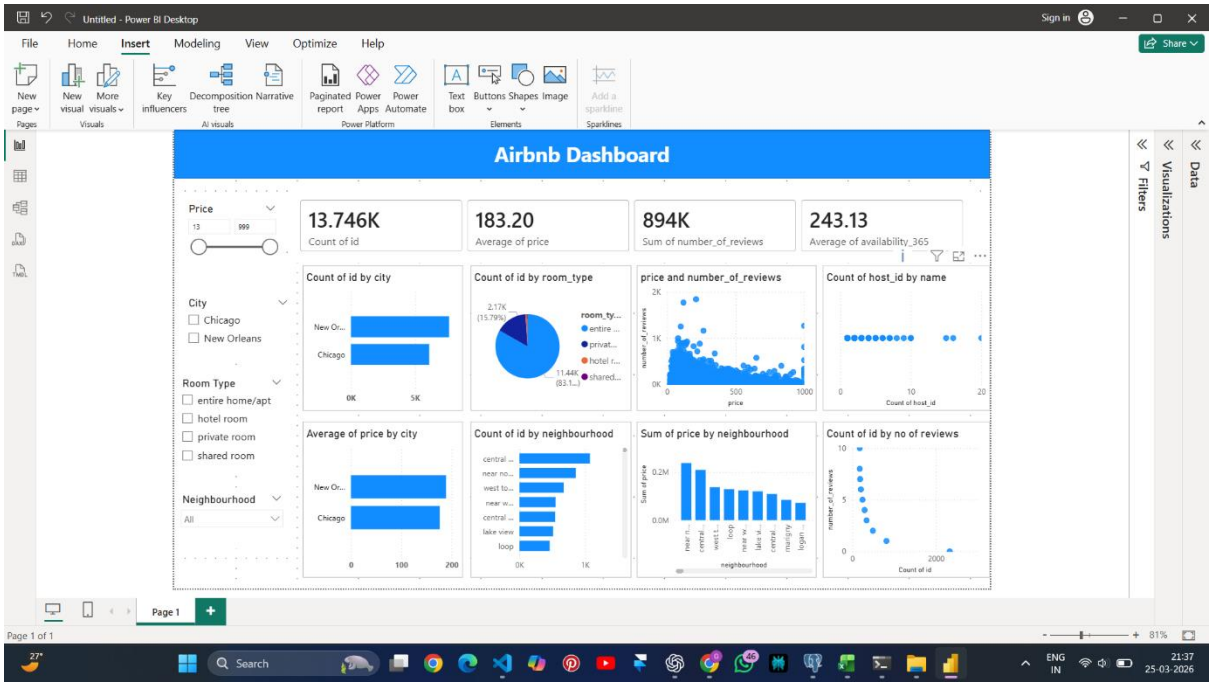
- **Top Hosts:** A few professional hosts manage many listings and dominate the market.
- **Reviews vs Host Type:** Hosts with more listings receive more total reviews, but average reviews per listing are similar.
- **Pricing vs Host Type:** Professional hosts charge slightly higher than individual hosts.
- **Insight:** Super-hosts dominate activity, and host type influences pricing and total reviews.

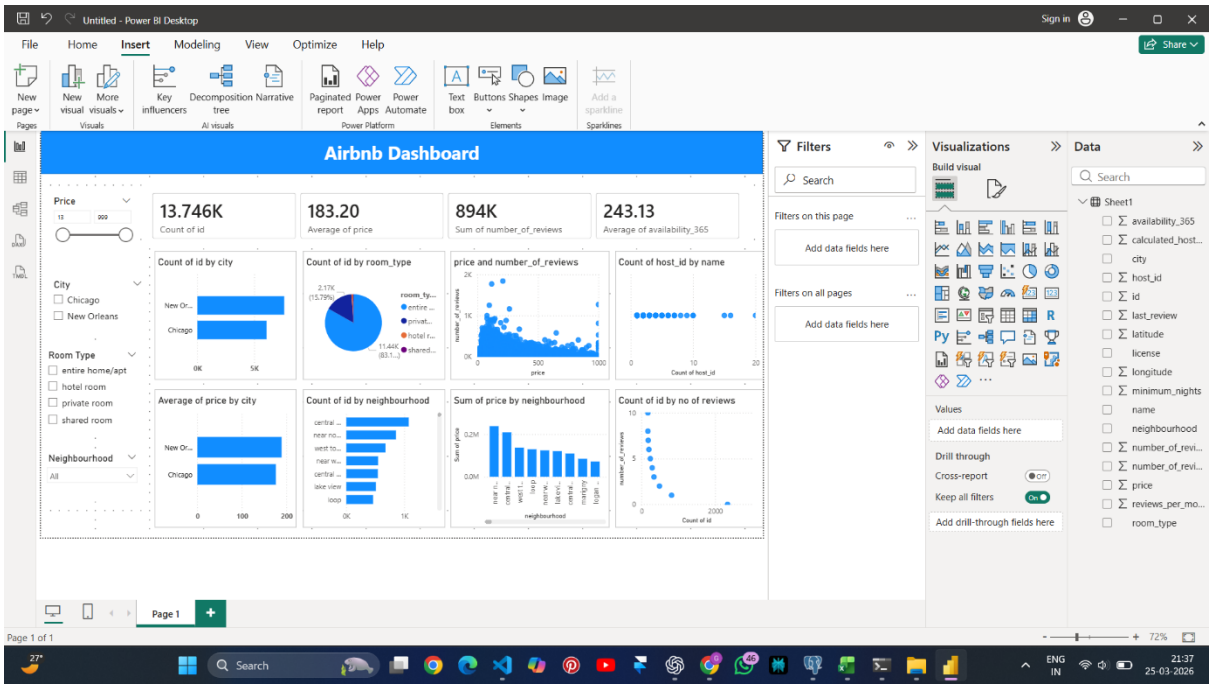
Dashboard Layout (Power BI)

Sections & Visuals:

1. **Overview:** KPI Cards – Total Listings, Avg Price, Total Reviews, Avg Availability; Bar Charts – Listings by City, Avg Price by City
2. **Property Analysis:** Pie/Donut Chart – Room Type Distribution; Bar Charts – Property Type Distribution, Top Neighbourhoods
3. **Pricing Analysis:** Bar Chart – Avg Price by Room Type; Scatter Chart – Price vs Reviews; Table/Bar Chart – Price by Neighbourhood
4. **Host Analysis:** Bar Chart – Top Hosts; Histogram/Bar Chart – Listings per Host; Optional Scatter – Reviews vs Listings per Host
5. **Filters/Slicers:** City, Room Type, Neighbourhood, Price Range

Dashboard Images





Conclusion

- Chicago dominates in total listings; New Orleans shows higher average prices in key areas.

- Entire homes/apartments dominate listings; room and property type preferences differ by city.
- Central locations and popular listings are priced higher.
- Professional hosts dominate the market and influence both pricing and total reviews.

Recommendation:
Airbnb can focus on **supporting professional hosts** and **optimizing pricing strategies** for central and high-demand neighbourhoods. This dashboard allows management to **track performance**, **identify trends**, and **make data-driven decisions**.